An Integrated TFT–MPC Framework for Optimal ESS and LNG Dispatch under Grid Congestion: A Case Study of the Honam Region

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Abstract

The Honam region, despite its concentration of low-carbon generation resources, experiences grid congestion due to limited interconnection line capacity to external grids. This leads to persistent curtailment under surplus power conditions, resulting in inefficient utilization of generation assets and challenges in the optimal operation of the energy mix. This study proposes an integrated operational framework that combines Temporal Fusion Transformer (TFT)-based 24-hour load forecasting with Rolling Horizon Model Predictive Control (MPC). The framework derives optimal dispatch strategies for Energy Storage Systems (ESS) and Liquefied Natural Gas (LNG) generation.
Case study results indicate that, in simulation, LNG generation decreases by approximately 2.9% compared to a rule-based baseline. Grid export increases by 16.5%, while daytime import decreases by 2.2%. The average daily operating cost is reduced by 8.17%.
These results demonstrate that the proposed framework enhances operational efficiency and enables more effective ESS-integrated dispatch for large-scale power system operation and renewable energy integration.

Introduction

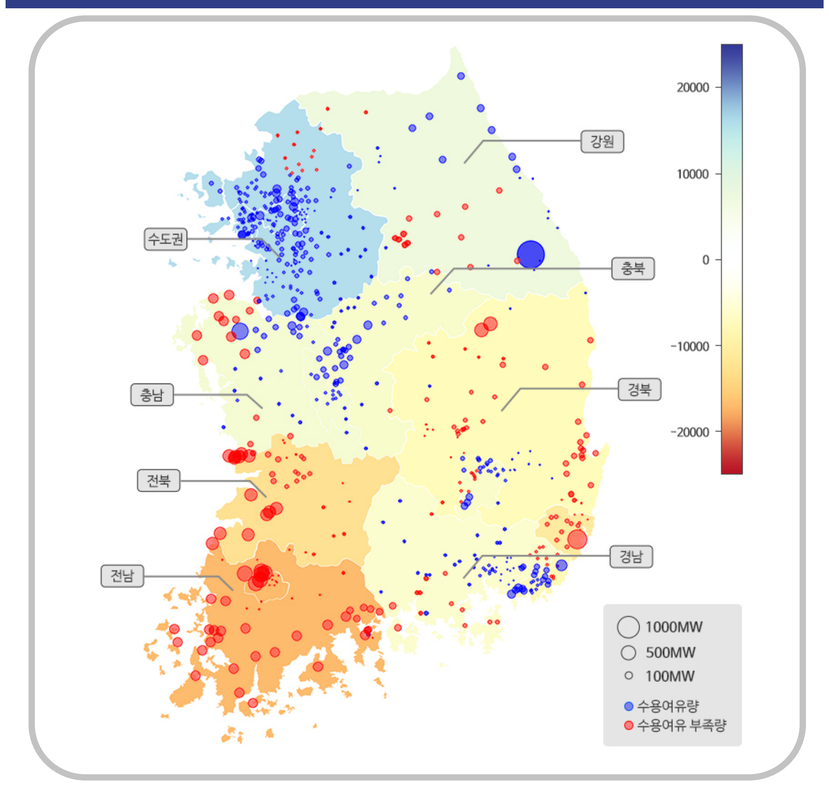


Figure 1. power system hosting capacity outlook for South Korea (2026-2030)

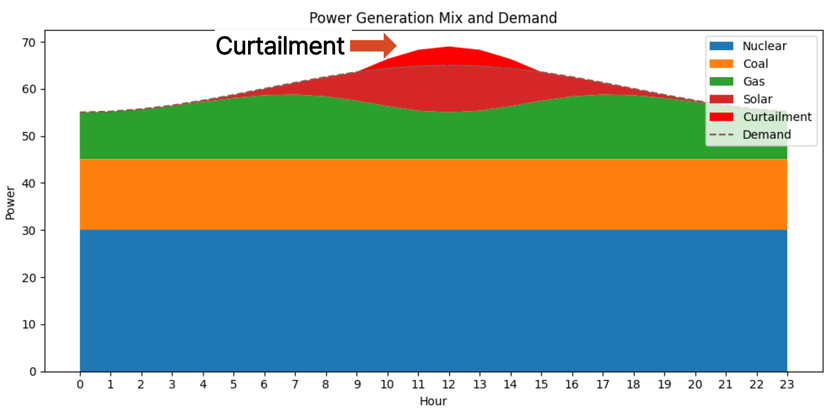


Figure 2. High solar generation during daytime creates surplus power that cannot be fully utilized, leading to curtailment. This indicates insufficient flexibility in the current system and the need for optimized control strategies.

Methodology

1. Overall pipline

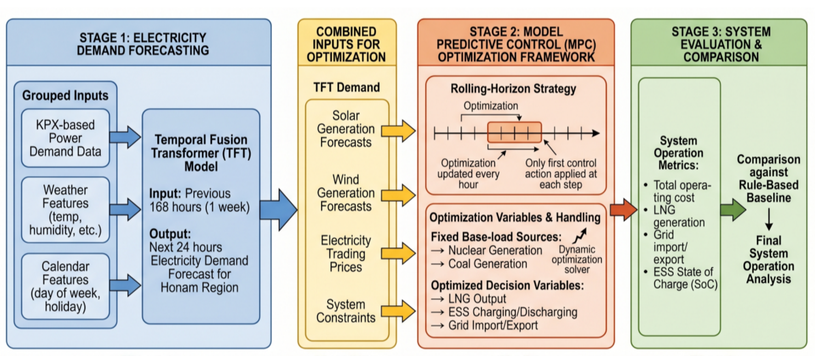


Figure 3. Overall pipeline of the proposed TFT-MPC framework for demand forecasting and power system operation optimization.

1. Overall pipline

Demand Forecasting Setup
- Inputs: demand, weather, calendar
- Input window: 168 h
- Forecast horizon: 24 h
- Model: TFT

MPC Optimization Setup
- Fixed sources: nuclear, coal
- Control variables: LNG, ESS, grid import/export
- Constraints: LNG limits, ramp rate, ESS SoC, grid limits
- Objective: minimize operating cost

Data description

The forecasting model was trained using hourly KPX demand data combined with weather and calendar features from 2014 to 2023, excluding the COVID-19 period (2020–2021), and tested on 2024 data.

2. TFT-based demand forecasting

	MAE(MWh)	RMSE(MWh)	MAPE(%)	R2
BiLSTM + Attn	482.3	613.0	5.504	0.6631
TFT	385.6	499.7	4.497	0.7761
VMD+CNN-LSTM	413.4	526.8	4.793	0.7512

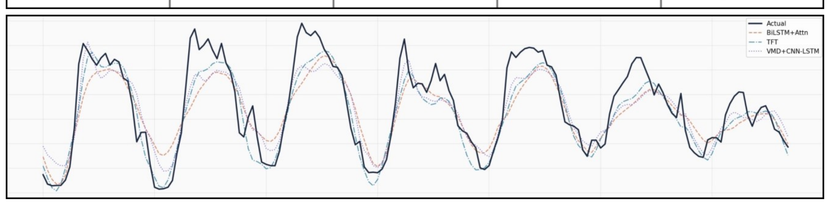


Table 1, Figure 4. Among the compared models, TFT achieved the best forecasting accuracy and tracked the actual demand pattern most closely. Therefore, TFT was selected as the demand forecasting model for the proposed framework.

3. Honam power system modeling

The Honam power system comprises fixed nuclear (4,300 MWh/h) and coal (1,000 MWh/h) baseload, controllable LNG (420–2,100 MWh/h), and a 5,600 MWh ESS (SoC: 10–90%, efficiency: 95%). Grid interconnection is capped at 3,000 MWh/h, with a tighter 1,600 MWh/h daytime import limit.

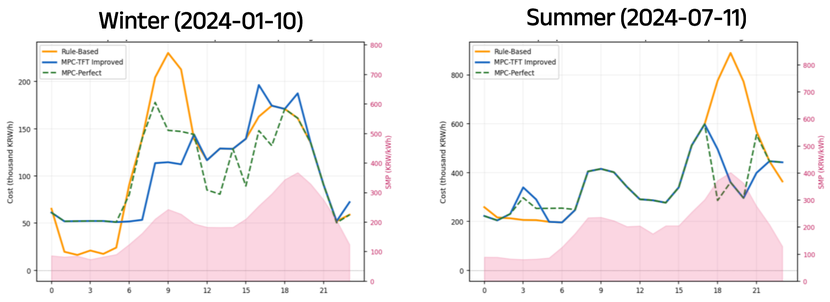
The System Marginal Price (SMP) follows a time-of-use profile (63–310 KRW/kWh) with seasonal multipliers (winter ×1.25, summer ×1.30) and ±8% daily stochastic variation, averaging ~176 KRW/kWh across the simulation period.

4. MPC-based operation optimization

MPC formulation
- **Inputs:** demand forecast, renewable forecast, electricity prices
- **Controls:** LNG, ESS, grid import/export
- **Objective:** minimize operating cost
- **Constraints:** LNG limits, ramp-rate, ESS SoC, terminal SoC, grid limits, power balance

$$\min \sum_{t} \left(\text{Cost}_{LNG} + \text{Cost}_{ESS} + \text{Cost}_{Import} - \text{Revenue}_{Export} \right)$$

The objective function minimizes the total operating cost, including LNG generation, ESS operation, electricity import cost, and export revenue over time.



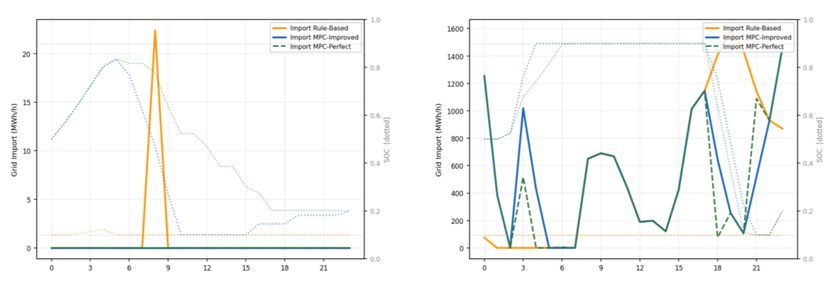


Figure 5. Sample-day comparison of hourly operating cost and grid import behavior for two representative days. MPC-TFT Improved closely tracks MPC-Perfect in both cost profile and ESS scheduling across seasons, with a cost gap of only **0.9%**, confirming that the TFT demand forecast captures **90.5%** of the theoretically achievable optimization gain under perfect foresight.

Results

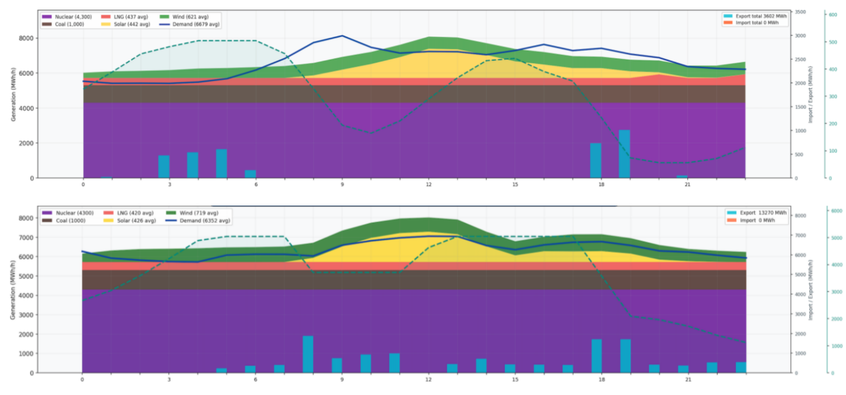


Figure 6. MPC-based operation adapts to seasonal conditions, reducing peak import and improving ESS utilization for cost-efficient system operation.

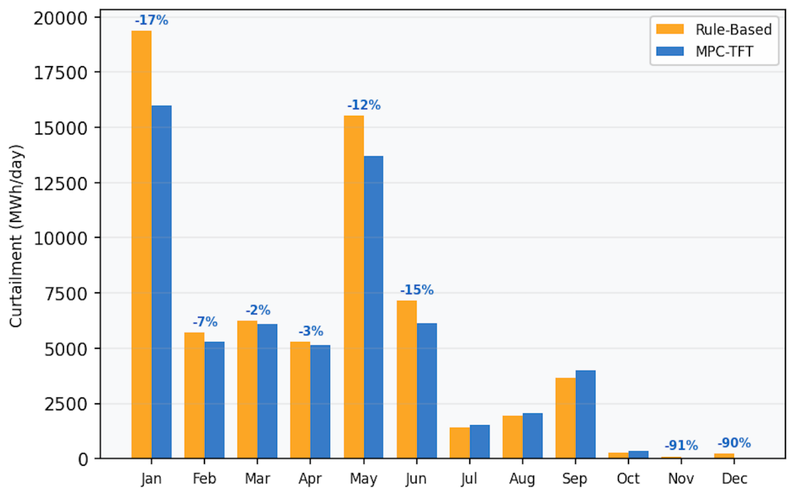


Figure 7. MPC-TFT reduces curtailment by up to 17% in high-surplus months (Jan, May, Jun) by pre-discharging ESS before renewable peaks, with an average reduction of 9.3%

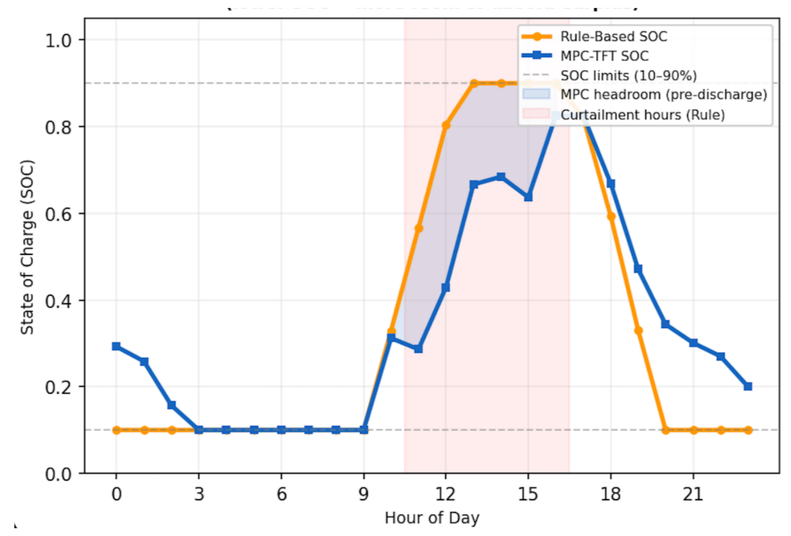


Figure 8. ESS SOC - MPC Pre-discharge Advantage (lower SOC → more room to absorb surplus

- Reduced total operating cost by 8.17% compared to rule-based operation
- Increased grid export by 16.5%, improving energy utilization
- Reduced daytime grid import by 2.2%, alleviating peak demand dependency
- Decreased LNG generation by 2.9% through optimized dispatch
- Reduced renewable curtailment by up to 17%, enhancing system efficiency

Discussion

In this study, nuclear generation is modeled as a fixed baseload to simplify the dispatch problem and highlight the role of flexible resources under grid constraints. However, this assumption does not capture the limited operational flexibility of nuclear plants. Future work may incorporate constrained nuclear flexibility, including ramping limits and xenon dynamics, using physics-informed or hybrid approaches that integrate operational data with physical constraints.

Reference

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